Deployment architecture

Customer-hosted control plane. People and agents use HTTPS. Only application nodes connect to PostgreSQL.

Operators reach the UI and the API. Agents call out to register, send a heartbeat, claim work, and post the result. The database is not on the public network and is not opened to worker hosts.

Components

Piece	Role
Control plane	Web UI, API, scheduler, roles, and workspaces. More than one node can share the database. One lease keeps a single scheduler.
PostgreSQL	Definitions, the queue, run history, and agent registration.
Agents	Workers on customer hosts. They run COMMAND, HTTP, and GIT_RELEASE jobs next to local files.
Identity	Local accounts or the customer's identity provider. Remote agents use a short-lived token from that provider.

Supported footprints

Shape	Typical use	Notes
Docker Compose	Production starter	PostgreSQL, control plane, and agents. A second application node is available for a higher-availability layout.
Kubernetes	Cluster production	Helm chart for the control plane, agents, and optional in-cluster Postgres.
Bare metal or VM	Traditional estates	Java 21 for the control plane. The same agent package on any JVM host.
Local JAR	Sandbox only	Embedded database. Not the production pattern.

Where a job runs

- **Agent mode** is the production path. Workers claim queued jobs. Scripts stay on the agent host.
- No pool and no preferred agent: any online agent with free capacity may take the job.
- A workspace pool and no preferred agent: any online member of that pool.
- A preferred agent: only that name. The job waits if that host is down or full.
- An empty pool claims nothing. Those jobs stay queued until agents join it.
- **Local mode** runs the job inside the control plane. Use it for a lab, not as the federated production path.

Network and operations

- People use HTTPS to the control plane.
- Agents use outbound HTTPS only. Remote agents present a short-lived token. Private-network agents may use a local operator login.
- Application nodes use a private connection to PostgreSQL.
- Restrict the UI to a private network where policy requires it.
- Back up PostgreSQL. It is the durability boundary.
- Use a separate control plane for pre-production when policy requires it. Move definitions with export, review, and apply. The file carries workspace and pool names, not hostnames or run history.
- Health and metrics are on the application tier.